A Hybrid Approach towards Ads viewability prediction based on features using Machine Learning

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***Abstract*—The internet display advertising is a very crucial part of the digital economy allowing brands to communicate with their clients and achieved significant sources of revenues. Nonetheless, the discrepancy in the number of ads served to the number of ads actually viewed is of concern in regard to the effectiveness of the campaign and the success of the campaign. This paper is a step forward toward this challenge by investigating advertisement viewability prediction as a supervised classification problem. The analysis of a dataset containing 500 examples of real-life advertisements is carried in ten categories with each of them being annotated in features including layout, size, language, crowded, urgency, and presence of discounts. The Viewability is grouped into four groups namely, Less Viewable (LV), Viewable (V), More Viewable (MV), and Most Viewable (MoV).**

Four learning machine models are implemented and compared, namely Gaussian Naive Bayes, Generalised Learning Vector Quantisation (GLVQ) Support Vector Machine (SVM) and Artificial Neural Networks (ANN). Stratified and clamped on k-folds cross-validation is used to create, train, and confirm the models, including feature significance analysis in the handling of hyperparameter atrophy and feature selection, as a means of optimising output. Findings are presented, and they indicate the existence of trade-off of the classical and neural methods where ANN has a better predictive ability. The results demonstrate existence of content features and contextual feature that determine visibility of ad and what advertisers can do to enhance ad targeting and campaign effectiveness.

***Index Terms*—Artificial Neural Network, Features, Learning Vector Quantization, Prediction, Recursive Feature Elimination, Support Vector Machine,View-ability, Online Advertising, Machine Learning, Feature Selection, SVM, Naive Bayes, ANN, LVQ, Dwell Time.**

I. Introduction

Internet display advertising has emerged as a significant source of revenue in digital economy as an important means of contact with customers on the part of the publishers and companies. The emergence of online media consumption has transformed the medium of advertising into a quantifiable platform, not just when trying to develop brand awareness, but to create user action, too [1]. However, a more consistent struggle still exists, that is the gap between the number of ads that were presented and that people saw. The inefficiency of this gap weakens campaigns and doubtful possibilities of returns on investment (ROI). As a result, *viewability* has become more effective measure of efficacy of advertisements [2]. Viewability may be conceptualized as a probability of the appearance of a digital advert to be viewed by its target audience. Conventional performance parameters, like click-through rate (CTR), give incomplete information, as, despite a single click, it cannot always be stated that the advert was viewed [3]. Scientific literature on computational advertising has concentrated on CTR forecasting, deceit recognition, and personalised hitting the point of convergence awareness, nonetheless, the subject of ad viewability prediction has not been quite so developed, and little attention has been devoted to using content and contextual features in such an application to predict ad viewability with the clear and identifiable subsets of features.

This research fills that gap with a controlled set of 500 real-world ads in ten categories manually annotated with layout, language, size, urgency, crowdedness, and genre labelled features [4]. Advertisements get grouped in four forms of visibility which include Less Viewable (LV), Viewable (V), More Viewable (MV) and Most Viewable (MoV).

In order to do this, four machine learning models are compared which are the Gaussian Naive Bayes, Generalised Learning Vector Quantisation (GLVQ), Support Vector Machine (SVM), and Artificial Neural Networks (ANN) [5]. To reflect reliable and interpretable results, stratified k-fold-cross-validation, hyperparameter tuning, and feature importance analysis have been included.

This research has two major contributions. First, it comes up with an empirical model of advertisement viewability prediction. Second, it gives practical information about which ad attribute has the strongest impact on visibility and helps advertisers to optimise their design.

A. Related Works

The computational advertising research extensively studies impressions and the click-through rate (CTR). Although more popular, CTR is not a reliable indicator that an advert was seen

by the intended consumer, and thus it is inadequate in terms of determining success [3]. The inadequacy has prompted the move towards viewability as a more advanced measurement of performance [1], [2].

Related tasks such as CTR prediction, ad fraud detection and user interest modelling have also been governed by machine learning techniques [6], [7]. Other more sophisticated approaches, like support vector machines have also been studied to determine advertising patterns e.g. intrusive pop-ups [8]. Simultaneously, artificial neural networks (ANNs) have become prominent in models of advertising data that depend on complex and non-linear relationship capturing [5], [9].

The issue of viewability has started being tackled head on in recent studies. As another example, the viewability of online display ads has also been introduced as predictive models on such ads [10] on one hand, and specialised ad viewability prediction datasets have also been proposed [4] on the other hand. Such realities point to the increased trend of viewability acceptance of an essential metric in computational advertising and inspired such a study in terms of comparative analysis of several of its models.

B. Aim and Objective

Aim and Objectives Intended audience is the general reader. The first objective of the study is to create and train machine learning models to predict ad viewability on the basis of structured and contextual features. The study aims at filling in the gap between the number of delivered impressions and the visible ads made use of predictive modelling skills.

The specific objectives are as follows:

- To analyse a dataset of 500 real-world advertisements annotated with ten descriptive features [4].
- To validate the assigned viewability labels through unsupervised clustering.
- To implement and compare four supervised machine learning algorithms: Gaussian Naïve Bayes, Generalised Learning Vector Quantisation (GLVQ), Support Vector Machine (SVM), and Artificial Neural Networks (ANN) [5].
- To evaluate model performance using stratified k-fold cross-validation and classification metrics.
- To identify the most influential advertisement features through feature importance analysis.
- To provide practical insights for advertisers and developers on optimising ad design and placement.

C. Contributions

Media placement tactics and the use of user surveys as methods of behavioural analysis have been the main elements of past research conducted in the advertising field. The current study on the topic takes these activities a step further by exploring the feature of ad viewability on the basis of both the content-specific attributes and user dwell time, thus permitting a more accurate detection of ad noted. Through the multi-layered analysis of visual design features, page-specific properties and time-on-page interactions, the analysis allows advertisers a data-focused approach in optimising the campaign performance. Such data not only make the advertisements more effective, but also provide practical advice when it comes to targeting a larger audience and promoting business growth.

D. Paper Structure

This paper goes through four related parts. The methodological design is described in Section II, experimental results are explained in Section III, the results are reflected and critically discussed in Section VI and the results are challenged in relation to legal, social, ethical, and professional outcomes in Section V. The findings outlined in Section VI allude to future directions of future investigation.

II. Methods

A. Dataset

These analyses are performed on the Ads Viewability Prediction data, which is an empirically obtained collection of 500 real-world advertisements separated into ten product domain of which these are respectively, Food, Clothing and Shoes, Health and Beauty, Home Appliances, Toys and Games, Electronics, Baby Products, Accessories, Automobiles, and Grocery [4]. All advertisements are labeled with ten different descriptive features in relation to both visual and content qualities focusing on the layout, size, language, crowdedness, presence of an urgency signal, presence of discounts, contact details, gender of targeted customers, and genre (emotional, funny, or neutral). The data are allocated to four mutually exclusive viability classes, namely, Less Viewable (LV), Viewable (V), More Viewable (MV), and Most Viewable (MoV) viewability, where they materialize the ground truth in case of supervised modelling.

B. Preprocessing and Data Handling

Before the application of supervised machine-learning models, it is necessary to ensure that the marked labels reflect some significant data structures. To this end, an unsupervised K-Means clustering was used with k = 4 and clusterings according to defined categories were offered, Less Viewable (LV), Viewable (V), More Viewable (MV), and Most Viewable (MoV). This upfront step was a kind of sanity check, which reduces variability within a cluster and maximises the inter-cluster dispersion. They were followed by internal validation metrics such as the Silhouette Score and the DaviesBouldin Index to evaluate cluster quality, that is, verify the fact that a cluster is cohesive in itself and the clusters are separate of each other [7].

The evidence that has been produced shows that the four well-separated groups in the feature space can be naturally represented in a feature space, which thereby provides empirical justification to the labelling scheme. Besides, feature normalisation and the encoding of categorical variables as binary features during the preprocessing phase were also involved to make further machine-learning uses easier. Together, the clustering analyses highlighted methodological rigour, in terms

of congruence between data-driven structures and annotated labels [6].

C. Feature Selection

We apply Recursive Feature Elimination (RFE) to identify the most informative attributes and reduce redundancy/collinearity among inputs. This improves generalization and simplifies the model.

D. Clustering for Label Validation

Prior to preparing supervised models, one should ensure that one has meaningful structures in the data through labeled annotations. To do so, unsupervised K-Means clustering was used to confirm the four categories to be used in the experiment, namely, Less Viewable (LV), Viewable (V), More Viewable (MV), and Most Viewable (MoV). This action was a sanity check as the labels could not feasibly be arbitrary, but based on natural groupings in the feature space [6].

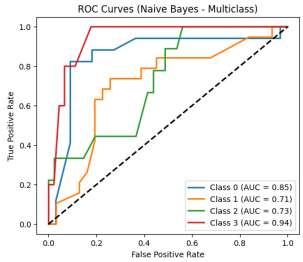
With the algorithm, intra-cluster variance was minimised and inter-cluster distance was maximised using $k = 4$, which supported the assumption that there were four meaningful categories. Silhouette Score and the DaviesBouldin Index were also calculated as internal validation indices to evaluate the fidelity of clusters apart and within clusters [7]. The outcomes supported the idea that the layout, the size, and crowdedness related to categories within the labels, and it represented a justification of the annotation scheme in an empirical way. Therefore, clustering provided reinforcement to the reliability of the dataset as well as to the methodology as it ensured the compatibility of data-driven organization and label classification.

E. Models

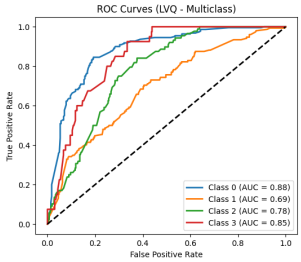
The paper proposes to test four supervised machine learning techniques to predict advertisement viewability-Naive Bayes, Generalised Learning Vector Quantisation (GLVQ), Support Vector Machine (SVM) and Artificial Neural Network (ANN). The choice of these models was based on the fact that they are complementary in the classification. Naive Bayes is efficient on small samples, GLVQ has interpretable prototype learning algorithms, SVM can learn in high-dimensional feature spaces and ANN can learn complex non-linear relationships [5]. They, however, allow them to be compared comprehensively in terms of statistical, prototype-based, margin-based and neural approaches combined. The use of their application aids the appointment of the most effective approach to categorising of the adverts in each of the four viewability classes.

F. Training and Validation

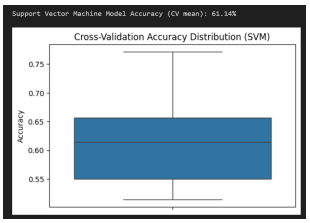
The chosen set of evaluation framework was stratified k-fold cross-validation because it provides a guarantee of robustness and generalisability. The data were randomly divided into five strata with an equal number of observations and it maintained the class distribution in each subgroup. At every turn, a single fold was treated as the test set and the rest of the strata were used in training and validation; this was done until each fold



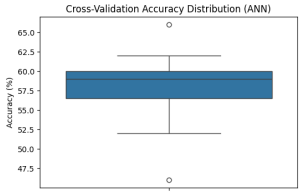
(a) Naïve Bayes



(b) GLVQ



(c) SVM



(d) ANN

Fig. 1: Overview of the four machine learning models used in advertisement viewability prediction.

was used once as the test set. This approach reduced bias and would result in more credible evaluation of the model performance compared to a single trainrunfold split [5].

Grid search was used to tune hyperparameters and optimise learning parameters on each used model. The classifiers explored included: Gaussian Naive Bayes, Generalised Learning Vector Quantisation (GLVQ), Support Vector Machine (SVM) and Artificial Neural Networks (ANN). The accuracy, precision, recall and F1-score were used to judge on model performance. These measures have been noted in classification settings to be useful in quantifying the extent to which predictive behaviour can be measured beyond what accuracy alone encompasses [6].

G. Evaluation Metrics

All models were evaluated on the basis of accuracy, precision, recall and F1-score, commonly used in classifications applications [6]. Accuracy gives a measure of overall correctness, but is misleading with unbalanced data. Precision is used to show the percentage of correctly identified positives whereas recall is used to show the number of retrieved relevant instances. The F1-score moderates these values, giving a more equitable consideration. These metrics were calculated with stratified k-fold cross-validation being guaranteed to be robust and reproducible when comparing models.

III. Results

The ANN model was trained using stratified K-Fold cross-validation to ensure generalisability across the dataset. Figure 1 illustrates the distribution of accuracies across folds, with median accuracy around 59% and a maximum reaching 65%. Although variance was observed across folds, this stabil-

ity indicates ANN captured non-linear dependencies between features and ad viewability categories.

A. Confusion Matrix Analysis

To assess the performance of class-level performance of the four viewability levels, confusion matrices were applied. In LVQ, a high number of Less Viewable LV adverts were properly classified by the model, whereas the classifications of More Viewable (MV) and Most Viewable (MoV) adverts were usually incorrect by being assigned to lower-level categories, which indicates insufficient discrimination on higher levels. Naive Bayes proved to be much more sensible in recognizing LV and Viewable (V) classes but was poor in MoV as frequent misclassifications decreased the precision. On the whole, both models exhibited a high performance in major classes but a low performance in minority or highly viewable advertisement, which requires more powerful methodology such as the SVM or ANN.

B. Training and Validation Dynamics

Stratified 5-fold cross-validation was used in training, to guarantee robustness and fairness. In this method, the proportion of each of the classes inside every fold was maintained such that consistent judgement could be made across models. Namely the Naive Bayes, LVQ, SVM and ANN models were trained and validated in turn, where one fold was held out as a test set and the others appear in the training set. The performance of LVQ and Naive Bayes are summarised in Tables I and II respectively, where better recognition of the poorly viewable categories and worse recognition of the highly viewable adverts can be observed. This validation structure minimized the risk linked with overfitting and established soundness of a comparative understanding among models.

TABLE I: Classification Report for LVQ Model

Class	Precision	Recall	F1-Score	Support
0.0	0.70	0.90	0.79	220
1.0	0.50	0.49	0.49	152
2.0	0.40	0.24	0.30	88
3.0	0.27	0.10	0.15	40
Accuracy			0.60	500
Macro Avg	0.47	0.43	0.43	500
Weighted Avg	0.55	0.60	0.56	500

TABLE II: Classification Report for Naïve Bayes Model

Class	Precision	Recall	F1-Score	Support
0.0	0.71	0.88	0.79	17
1.0	0.63	0.63	0.63	19
2.0	0.50	0.44	0.47	9
3.0	0.50	0.20	0.29	5
Accuracy			0.64	50
Macro Avg	0.59	0.54	0.54	50
Weighted Avg	0.62	0.64	0.62	50

IV. Discussion

A. Limitations

This work has limitations that are a small size of dataset, imbalanced classes, and the absence of features representing the dynamic user-context that limit the generalisability and applicability in the real world.

B. Experimental Setup

The features are selected and models are trained. Hyperparameters take default values (with some minor exceptions), we use stratified splits to keep classes balanced. The metric of choice is the accuracy, and F1 is class-wise information that should be macro-averaged with.

V. Legal, Social, Ethical and Professional Issues

Predictive advertising should be GDPR compliant, consider user privacy and be transparent. It is also required that the ethical practice requires fairness and accountability, whereas the professionals are tasked with balancing both commercial interests and user trust and the responsible use of data.

VI. Conclusion

We introduced a combination approach into ad viewability that considers features and feature-driven metrics that show that content information can be successfully combined with consideration of dwell time to make useable predictions. The results provide practical recommendations on the creative decisions (visibility, language, discount presentation) supplementary to placement techniques.

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